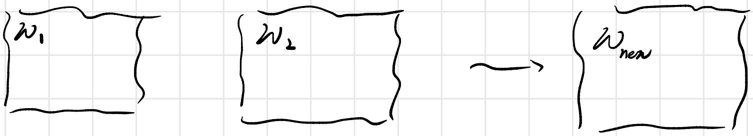
To merge two models, we sometimes need to take two matrices (one from each) and combine them into one matrix.



note we require them to have the same shape,
otherwise we could do dimensionality reduction first, e.g. PCA

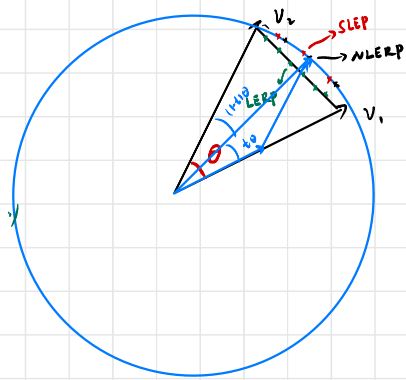
method 1: LERP (linear extrapolation)

$$V = \alpha V_1 + (1-\alpha) V_2, \quad 0 \le \alpha \le 1$$

problem: shrinkage

$\|V\|_2^2$ smaller than $\|V_1\|_2^2$ or $\|V_2\|_2^2$

(generally should be around 1)



method 2 NLERP
(normalized LERP)

$$V = \frac{\alpha V_1 + (1-\alpha) V_2}{\|\alpha V_1 + (1-\alpha) V_2\|_2}$$ → cares about direction only, norm = 1

$$L = \alpha \|V_1\|_2 + (1-\alpha)\|V_2\|_2$$ → care about norm only

$$V \leftarrow V \cdot L$$ → rescale

method 3 SLERP (spherical linear interpolation)

$$V_1^* \leftarrow \frac{V_1}{\|V_1\|_2}, \quad V_2^* \leftarrow \frac{V_2}{\|V_2\|_2}$$ scale first, so V_1, V_2 sit on the same circle

→ note we can show $\|V\|_2^2 = 1$

$$V \leftarrow \underbrace{\frac{\sin(\alpha\theta)}{\sin(\theta)} V_1^* + \frac{\sin((1-\alpha)\theta)}{\sin(\theta)} V_2^*}, \quad \text{where } \cos\theta = \frac{V_1^* V_2^*}{\|V_1^*\|_2 \|V_2^*\|_2}$$

→ change $\alpha \in [0,1]$
V travels between V_1 and V_2 through the circle evenly

$$L \leftarrow \underbrace{\alpha \|V_1\|_2 + (1-\alpha)\|V_2\|_2}$$

$$V \leftarrow V \cdot L$$ → now rescale v

Compare SLERP and NLERP

SLERP: $t: 0 \to 1$ move evenly along the circle

NLERP: move unevenly